Customer Churn Prediction – Group 6
University Canada West
BUSI 651-01

```
Logistic Regression Results:
Accuracy: 0.6850

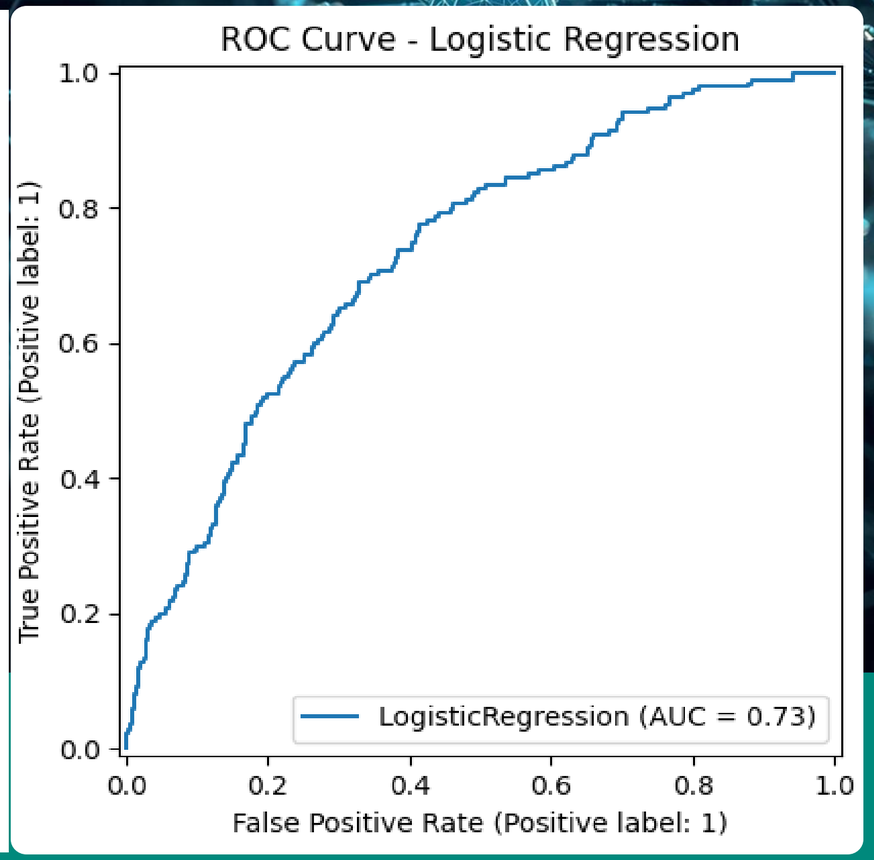
Confusion Matrix:
 [[699 314]
 [ 64 123]]

Classification Report:
              precision    recall  f1-score   support

           0       0.92      0.69      0.79      1013
           1       0.28      0.66      0.39       187

    accuracy                           0.69      1200
   macro avg       0.60      0.67      0.59      1200
weighted avg       0.82      0.69      0.73      1200


Top 3 Factors Increasing Churn Risk:
Contract_Month-to-month           0.321684
PaymentMethod_Electronic check    0.281359
SeniorCitizen                     0.167397
dtype: float64

Top 3 Factors Reducing Churn Risk (Retention Drivers):
Contract_Two year                      -0.577641
OnlineSecurity_Yes                     -0.261563
PaymentMethod_Credit card (automatic)  -0.251721
dtype: float64
```

```
--- Results with Adjusted Threshold (0.7) ---
New Accuracy: 0.8117
New Confusion Matrix:
 [[920  93]
 [133  54]]

Classification Report:
              precision    recall  f1-score   support

           0       0.87      0.91      0.89      1013
           1       0.37      0.29      0.32       187

    accuracy                           0.81      1200
   macro avg       0.62      0.60      0.61      1200
weighted avg       0.79      0.81      0.80      1200


Top 3 Factors Increasing Churn Risk:
Contract_Month-to-month           0.321684
PaymentMethod_Electronic check    0.281359
SeniorCitizen                     0.167397
dtype: float64

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